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```
#include <iostream>
#include <vector>
#include <omp.h>
#include <cstdlib>
#include <climits> // Include for INT_MAX, INT_MIN
#include <chrono>

using namespace std;
using namespace std::chrono;

int main() {
   int n = 1000000; // Size of array
   vector<int> arr(n);

   // Generate random numbers
   #pragma omp parallel for
   for (int i = 0; i < n; i++) {
      arr[i] = rand() % 1000; // Random values between 0-999
   }

   int minVal = INT_MAX;
   int maxVal = INT_MIN;
   long long sum = 0; // Use long long for large sums
   double avg = 0.0;

   // **Start Timing**
   auto start = high_resolution_clock::now();

   // **Parallel Reduction**
```

```
    #pragma omp parallel for reduction(min:minVal) reduction(max:maxVal) reduction(+:sum)
    for (int i = 0; i < n; i++) {
        minVal = min(minVal, arr[i]); // Corrected syntax
        maxVal = max(maxVal, arr[i]); // Corrected syntax
        sum += arr[i];
    }

    // **Compute Average**
    avg = static_cast<double>(sum) / n;

    // **Stop Timing**
    auto stop = high_resolution_clock::now();
    auto duration = duration_cast<milliseconds>(stop - start);
    // **Print Results**
    cout << "Parallel Reduction Results:\n";
    cout << "Min: " << minVal << endl;
    cout << "Max: " << maxVal << endl;
    cout << "Sum: " << sum << endl;
    cout << "Average: " << avg << endl;
    cout << "Execution Time: " << duration.count() << " ms\n";
    return 0;
}
```

O/P:

Parallel Reduction Results:

Min: 0

Max: 999

Sum: 496875044

Average: 496.875

Execution Time: 38 ms